Great Circles as Geodesics of the Sphere

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of great circles as geodesics of the sphere, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

DG-02 / Differential Geometry / Report v1

I. TARGET AND IMPORT CONTRACT

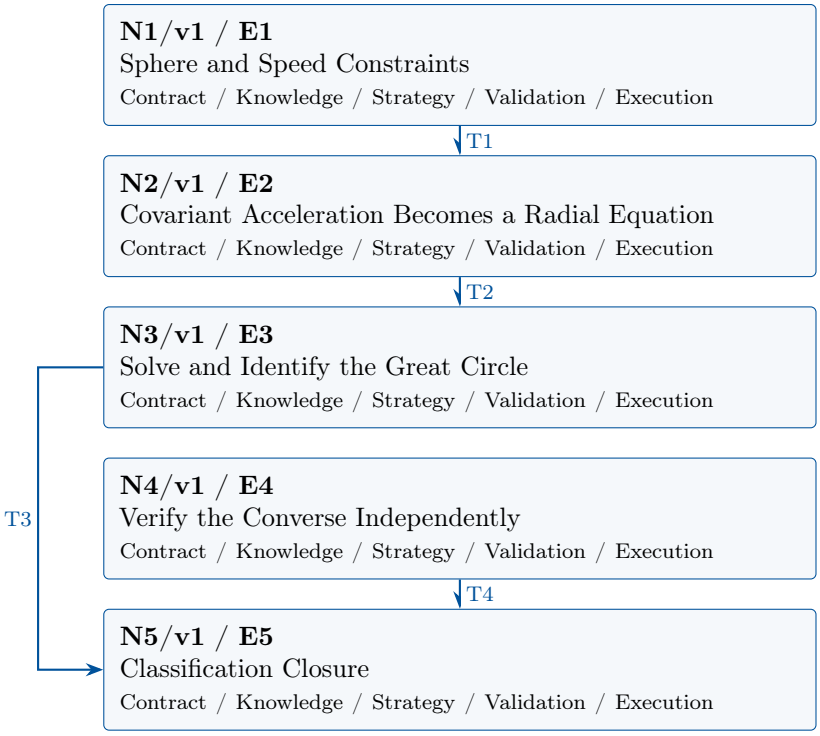
Theorem I.1. *On the round sphere $S^2_R \subset \mathbb{R}^3$, $R > 0$, every nonconstant unit-speed geodesic traces a great circle. Conversely every unit-speed great-circle arc is a geodesic.*

A great circle is the sphere's intersection with a two-dimensional plane through the origin. A geodesic has zero covariant acceleration. This is an intrinsic condition: its ordinary three-dimensional acceleration need not be zero.

A. Foundations and permitted imports

for the induced metric, the Levi-Civita covariant derivative is the tangential projection of the ambient derivative; elementary linear ODE uniqueness; inner-product differentiation. Curves are C^2, parameterized by arc length s. Global shortest-path claims are outside this theorem.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. $\langle \gamma, \gamma' \rangle = 0$ and $\langle \gamma, \gamma'' \rangle = -1$.

Receiving activity. N2: Covariant Acceleration Becomes a Radial Equation

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and the tangential-projection formula for the induced connection.

T2: N2/E2/v1 to N3. $\gamma'' = -\gamma/R^2$.

Receiving activity. N3: Solve and Identify the Great Circle

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, $\|p\| = R$, $\|v\| = 1$, $p \cdot v = 0$, and linear ODE uniqueness.

T3: N3/E3/v1 to N5. $\gamma(s) = p\cos(s/R) + Rv\sin(s/R)$, contained in the origin plane spanned by p, v.

Receiving activity. N5: Classification Closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and N4/E4.

T4: N4/E4/v1 to N5. Every unit-speed great-circle parametrization is a geodesic.

Receiving activity. N5: Classification Closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and N4/E4.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Sphere and Speed Constraints

Object and role. Sphere and Speed Constraints. Establish the following local result: $\langle \gamma, \gamma' \rangle = 0$ and $\langle \gamma, \gamma'' \rangle = -1$.

Inputs. A C^2 unit-speed geodesic on the sphere of radius $R > 0$.

C — Contract

Own E1/v1: $\langle \gamma, \gamma' \rangle = 0$ and $\langle \gamma, \gamma'' \rangle = -1$.

K — Knowledge

Round metric and arc length

S — Strategy

Fix point and tangent data

V — Validation

Require $R > 0$ and unit speed

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Differentiate $\langle \gamma, \gamma \rangle = R^2$ once and twice.

N1.2 Use $\|\gamma'\|^2 = 1$.

N1.3 Retain both tangent and acceleration constraints along the parameter interval.

Export E1/v1. $\langle \gamma, \gamma' \rangle = 0$ and $\langle \gamma, \gamma'' \rangle = -1$.

Acceptance gate. Require $R > 0$ and unit speed Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Require $R > 0$ and unit speed Recheck dependent nodes N2, N3, N5.

B. N2/v1: Covariant Acceleration Becomes a Radial Equation

Object and role. Covariant Acceleration Becomes a Radial Equation. Establish the following local result: $\gamma'' = -\gamma/R^2$.

Inputs. N1/E1 and the tangential-projection formula for the induced connection.

C — Contract

Own E2/v1: $\gamma'' = -\gamma/R^2$.

K — Knowledge

Tangential-projection connection

S — Strategy

Use normal-line characterization

V — Validation

Separate ambient and intrinsic derivatives

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Zero covariant acceleration makes ambient acceleration normal.

N2.2 The normal line is spanned by γ.

N2.3 Pair $\gamma'' = \lambda\gamma$ with γ to obtain $\lambda = -1/R^2$.

Export E2/v1. $\gamma'' = -\gamma/R^2$.

Acceptance gate. Separate ambient and intrinsic derivatives Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Separate ambient and intrinsic derivatives Recheck dependent nodes N3, N5.

C. N3/v1: Solve and Identify the Great Circle

Object and role. Solve and Identify the Great Circle. Establish the following local result: $\gamma(s) = p\cos(s/R) + Rv\sin(s/R)$, contained in the origin plane spanned by p, v.

Inputs. N2/E2, $\|p\| = R$, $\|v\| = 1$, $p \cdot v = 0$, and linear ODE uniqueness.

C — Contract

Own E3/v1: $\gamma(s) = p\cos(s/R) + Rv\sin(s/R)$, contained in the origin plane spanned by p, v.

K — Knowledge

Linear ODE solution

S — Strategy

Solve oscillator

V — Validation

Respect initial constraints

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Solve the constant-coefficient oscillator with the initial point and tangent.

N3.2 Write $e_1 = p/R$ and $e_2 = v$, an orthonormal pair.

N3.3 Identify the intersection of their plane with the sphere.

Export E3/v1. $\gamma(s) = p\cos(s/R) + Rv\sin(s/R)$, contained in the origin plane spanned by p, v.

Acceptance gate. Respect initial constraints Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Respect initial constraints Recheck dependent nodes N5.

D. N4/v1: Verify the Converse Independently

Object and role. Verify the Converse Independently. Establish the following local result: Every unit-speed great-circle parametrization is a geodesic.

Inputs. Any orthonormal pair e_1, e_2 and the round sphere metric; no forward classification is assumed.

C — Contract

Own E4/v1: Every unit-speed great-circle parametrization is a geodesic.

K — Knowledge

Great-circle parametrization

S — Strategy

Compute tangential acceleration

V — Validation

Require orthonormal directions

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Differentiate $R(e_1 \cos(s/R) + e_2 \sin(s/R))$.

N4.2 Check sphere membership, unit speed, and acceleration $-\gamma/R^2$.

N4.3 The acceleration has zero tangential projection.

Export E4/v1. Every unit-speed great-circle parametrization is a geodesic.

Acceptance gate. Require orthonormal directions Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Require orthonormal directions Recheck dependent nodes N5.

E. N5/v1: Classification Closure

Object and role. Classification Closure. Establish the following local result: Unit-speed geodesics on the round sphere are exactly unit-speed great-circle arcs.

Inputs. N3/E3 and N4/E4.

C — Contract

Own E5/v1: Unit-speed geodesics on the round sphere are exactly unit-speed great-circle arcs.

K — Knowledge

Both proof branches

S — Strategy

Join necessity and converse

V — Validation

Avoid global minimizing overclaim

E — Execution

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N5.1 Combine necessity and sufficiency.

N5.2 Use ODE uniqueness for the initial-data classification.

N5.3 Separate this geodesic statement from the stronger and generally false claim that every whole arc is globally shortest.

Export E5/v1. Unit-speed geodesics on the round sphere are exactly unit-speed great-circle arcs.

Acceptance gate. Avoid global minimizing overclaim Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N5: Avoid global minimizing overclaim Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Sphere and Speed Constraints

Let $\gamma(s)$ be a unit-speed geodesic. The sphere and speed conditions are

$$\langle \gamma, \gamma \rangle = R^2, \qquad \langle \gamma', \gamma' \rangle = 1.$$

Differentiating the first once gives $\langle \gamma, \gamma' \rangle = 0$, so velocity is tangent. Differentiating again gives

$$\langle \gamma, \gamma'' \rangle = -\langle \gamma', \gamma' \rangle = -1.$$

These identities are valid along the whole curve.

B. N2 — Covariant Acceleration Becomes a Radial Equation

The geodesic condition says the tangential projection of gamma' ' is zero. The normal line at gamma is spanned by gamma itself. Hence $\gamma'' = \lambda(s)\gamma$. Taking inner products with gamma and using N1 gives $\lambda(s)R^2 = -1$, so

$$\gamma'' = -\frac{1}{R^2}\gamma.$$

This nonzero inward acceleration keeps a geodesic on the curved sphere while its intrinsic acceleration vanishes.

C. N3 — Solve and Identify the Great Circle

At $s = 0$ choose initial point p and unit tangent v, so $\|p\| = R$, $\|v\| = 1$, and $p \cdot v = 0$. The unique solution of N2's ODE with these data is

$$\gamma(s) = p\cos(s/R) + Rv\sin(s/R).$$

Its image stays in the fixed plane span (p,v), which passes through the origin. Writing $e_1 = p/R$ and $e_2 = v$ gives an orthonormal pair and

$$\gamma(s) = R\big(e_1\cos(s/R) + e_2\sin(s/R)\big).$$

It therefore traces a great-circle arc. Any other initial parameter value can be shifted to zero, so this covers every unit-speed geodesic locally and throughout its parameter interval by uniqueness.

D. N4 — Verify the Converse Independently

Given any orthonormal e_1, e_2, the last formula satisfies $\|\gamma\| = R$, ||gamma'|| = 1, and gamma' '= $-\gamma/R^2$. Its acceleration is normal to the sphere, so its tangential projection is zero. The curve is a geodesic. This check does not assume the forward classification.

E. N5 — Classification Closure

N3 proves necessity and N4 proves sufficiency. Initial data determine a unique parameterized geodesic, while reversing the tangent reverses direction along the same great circle. The theorem follows. □

□

V. WORKED ORCHESTRATION

On the unit sphere, $p = (1, 0, 0)$, $v = (0, 1, 0)$ gives

$$\gamma(s) = (\cos s, \sin s, 0),$$

the equator. Ambient acceleration is $-\gamma$, entirely normal, while speed is one.

Now consider a latitude with $0 < |c| < 1$. Let $\rho = sqrt(1 - c^2)$ and

$$\eta(s) = (\rho\cos(s/\rho), \rho\sin(s/\rho), c).$$

This is unit speed. Its ambient acceleration is

$$\eta'' = (-\cos(s/\rho)/\rho, -\sin(s/\rho)/\rho, 0).$$

Since eta $\cdot eta$' '= -1, its tangential component on the unit sphere is eta' '+eta. Its third coordinate is c, so this component is nonzero. The latitude is not a geodesic. When $c = 0$, $\rho = 1$ and the tangential acceleration vanishes, recovering the equator.

A. Boundary and hypothesis audit

Unit speed matters: for constant speed c_0 the equation becomes gamma' '= $-c_0^2\ \gamma/R^2$. Arbitrary reparameterizations may not satisfy the affine geodesic equation even when their images remain great circles. Constant geodesics are excluded by the nonconstant unit-speed contract.

A geodesic arc need not minimize length globally. On a unit great circle, an arc of length $3\pi/2$ has the same endpoints as the opposite arc of length $\pi/2$. Antipodal endpoints admit multiple semicircular geodesics. Neither fact contradicts the classification.

Export the normal-acceleration bridge, the initial-value formula, and the great-circle equivalence. These supply concrete local geodesic intuition for later completeness and curvature problems.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

A branch identifies all geodesics with straight lines because it replaces covariant acceleration by ordinary acceleration.

- **Node:** N2
- **State:** STALLED
- **Failing component:** Validation
- **Failed definition:** geodesic means zero tangential acceleration
- **Observed:** ambient acceleration incorrectly set to zero
- **Repair target:** connection interpretation
- **Blocked export:** straight-line classification

Knowledge retrieves the induced connection. Validation distinguishes the two derivatives. Strategy uses normal acceleration instead of zero ambient acceleration. Execution derives gamma''$= -\gamma/R^2$ and verifies the equator. Contract retains the classification target and replaces the invalid bridge.

B. Hypothesis boundary

Unit speed matters: for constant speed c_0 the equation becomes gamma''$= -c_0^2\ \gamma/R^2$. Arbitrary reparameterizations may not satisfy the affine geodesic equation even when their images remain great circles. Constant geodesics are excluded by the nonconstant unit-speed contract.

A geodesic arc need not minimize length globally. On a unit great circle, an arc of length $3\pi/2$ has the same endpoints as the opposite arc of length $\pi/2$. Antipodal endpoints admit multiple semicircular geodesics. Neither fact contradicts the classification.

Export the normal-acceleration bridge, the initial-value formula, and the great-circle equivalence. These supply concrete local geodesic intuition for later completeness and curvature problems.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Harvard differential geometry notes: induced connections and geodesics.